

Slide 1: Operating Systems — Lecture 14

Virtual Memory & Paging Architectures

- Course: CS 301 - Operating Systems Principles
- Instructor: Dept. of Computer Science & Systems
- Core Focus: Memory virtualization, page tables, and demand paging
- Objective: Eliminate external fragmentation and isolate process address spaces

Slide 2: Motivation for Virtual Memory

Physical Memory Limitations & Logical Address Spaces

- Physical RAM is finite and shared among concurrent processes.
- Process address spaces must remain protected and isolated from one another.
- Paging separates logical memory from physical frames, enabling non-contiguous allocation.
- Only portions of a binary need to reside in physical RAM for execution (Demand Paging).

Slide 3: Paging Mechanics & Address Translation

Virtual Page Numbers and Page Offsets

- Physical memory is partitioned into fixed-size blocks termed Frames.
- Logical memory is partitioned into identical-sized blocks termed Pages.
- Frame size == Page size (typically 4 KB = 4096 bytes = 2^{12} bytes, requiring 12 offset bits).
- Virtual address format: (Page Number [p], Page Offset [d]).
- A per-process Page Table translates logical page numbers into physical frame numbers.

Slide 4: Translation Lookaside Buffer (TLB)

Hardware Acceleration for Fast Memory Access

- Naive paging doubles physical memory accesses: 1 for page table + 1 for actual data.
- The TLB is an associative, high-speed on-chip hardware cache.
- TLB Hit: Frame resolved in ~1 CPU clock cycle.
- TLB Miss: Requires accessing page table in main memory, then updating the TLB.
- Effective Access Time (EAT) = $\text{Hit_Ratio} * (\text{TLB} + \text{Mem}) + (1 - \text{Hit_Ratio}) * (\text{TLB} + 2 * \text{Mem})$.

Slide 5: Demand Paging & Page Fault Traps

Handling Non-Resident Memory

- Page Table Entry (PTE) includes a valid-invalid bit (1 = in RAM, 0 = on disk).
- Accessing a page with valid bit = 0 triggers a CPU hardware Page Fault trap.
- The OS kernel handles the trap: finds a free frame, reads block from backing store into RAM.
- The OS updates the PTE to valid=1 and restarts the interrupted instruction.
- Page fault service latency is dominated by disk I/O (~milliseconds vs nanoseconds).

Slide 6: Page Replacement Policies

Evicting Victim Frames Under Memory Saturation

- When physical RAM has no free frames, a page replacement algorithm selects a victim frame.
- FIFO (First-In, First-Out): Simple queue. Susceptible to Belady's Anomaly.
- Belady's Anomaly: Adding more physical frames can counter-intuitively increase page faults.
- LRU (Least Recently Used): Replaces page unused for the longest time (Stack Algorithm).
- Optimal (OPT / MIN): Replaces page that will not be used for longest future duration.

Slide 7: Thrashing and The Working-Set Model

System Performance Collapse Prevention

- Thrashing occurs when processes spend more time paging than executing instructions.
- When active working sets exceed physical memory, page faults cascade and CPU utilization drops.
- The Working-Set Window Delta defines the set of pages referenced in recent delta time.
- If sum of working sets > total frames, OS long-term scheduler must suspend/swap out processes.